

18 September 2026

Editors

Software Testing, Verification and Reliability

Dear Editors,

Re: Research Paper submission, “From Skill-Level to Predicate-Level: Budgeted Regression Test Selection for Evolving LLM Agent Skills”

Please consider our manuscript for publication as a Research Paper in *Software Testing, Verification and Reliability*. The paper addresses a practical maintenance problem created by evolving reusable LLM-agent skills: when an expensive interactive test suite previously passed, which tests should be replayed first after a sparse conditional change?

We introduce PredicateImpact, a dynamic change-impact method that ranks tests using predicate-level execution coverage, together with an auditable native mutation benchmark. Across two disjoint, content-blind ALFWorld/ScienceWorld confirmation rosters, PredicateImpact detects every killable mutation within a three-test budget. Its conditional detection rate is approximately 44 percentage points above a strong random baseline that already knows the changed skill. Frozen subgroup gates are passed in both environments and for deletion, same-kind argument substitution, and adjacent-order swap. An exact predicate-coverage-set random ablation ties the proposed ranking, locating the gain in coverage granularity rather than an unsupported ordering advantage.

The manuscript fits the journal’s focus on software testing and evolution by extending regression-test selection and change-impact analysis to executable agent skills while retaining explicit mutation operators, fixed budgets, independent confirmation sets, and reproducible integrity checks. Source code, data, frozen protocols, analysis scripts, and the reproducibility bundle are publicly available at <https://github.com/ls680/predicate-impact-agent-skill-regression-testing>.

We confirm that this manuscript has not been published previously and is not under consideration by another journal. Both authors approved the submitted version, the author order, and the contribution statement, and agree to be accountable for the work. The authors declare no competing interests and no specific funding. Generative-AI assistance is disclosed in the manuscript.

Thank you for considering our work.

Sincerely,

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